ICT Class 7 - Formative Assessment 1

Formative Assessment (FA-1)

- Class 7

Class: Class 7

Assessment Type: Formative Assessment (FA-1)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Networking & Internet, Advanced Graphics

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 — 1 = 5 marks)

Choose the correct answer:

1. Which type of network connects devices within a single building?
    1. (a) WAN
    2. (b) LAN
    3. (c) MAN
    4. (d) PAN
2. Which protocol is used for sending emails?
    1. (a) FTP
    2. (b) HTTP
    3. (c) SMTP
    4. (d) TCP
3. What does DNS stand for?
    1. (a) Domain Name System
    2. (b) Data Network Service
    3. (c) Digital Name Server
    4. (d) Domain Network System
4. In GIMP, which tool is used to remove the background from an image?
    1. (a) Crop Tool
    2. (b) Fuzzy Select Tool
    3. (c) Paintbrush Tool
    4. (d) Text Tool
5. Which layer in GIMP allows you to edit parts of an image independently?
    1. (a) Background Layer
    2. (b) Alpha Layer
    3. (c) Active Layer
    4. (d) All layers allow independent editing

Section II: Short Answer Questions (5 — 2 = 10 marks)

Answer briefly:

1. What is the difference between LAN and WAN? Give one example of each.

2. Name any four Internet services commonly used today.
3. What is the purpose of a protocol in networking?
4. Define the term “layer” in image editing. Why are layers useful?
5. What is the difference between a raster image and a vector image?

Section III: Long Answer Questions (1 — 5 = 5 marks)

Answer in detail:

1. Explain the types of computer networks (LAN, MAN, WAN) with suitable examples and a diagram.

Part B: Hands-on Practical Lab Task (5 marks)

Complete the following tasks on the computer:

1. Open GIMP and create a new image (800—600 pixels). Add two layers: one for background (fill with a color) and one for drawing a simple shape. Save the file as “LayerDemo.xcf”. (2 marks)
2. Open a web browser and identify the protocol used when visiting “https://www.example.com”. List the steps to find the IP address of a website using the command prompt. (2 marks)
3. Save your GIMP file on the desktop with the correct filename. (1 mark)

Part C: Notebook Maintenance (5 marks)

Marks will be awarded for:

- Regularity in completing classwork (2 marks)
- Neatness of notes and diagrams (2 marks)
- Completion of all class activities (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters

Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: “What I learned” and “What was hard” (2 marks)
- Real-life connections: “Where I see this outside school” (1 mark)

ICT Class 7 - Formative Assessment 2

Formative Assessment (FA-2)

- Class 7

Class: Class 7

Assessment Type: Formative Assessment (FA-2)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Multimedia, Data Handling

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 — 1 = 5 marks)

Choose the correct answer:

1. Which software is commonly used for editing audio files?
   1. (a) GIMP
   2. (b) Audacity
   3. (c) Scratch
   4. (d) LibreOffice Calc
2. What is the full form of MP3?
   1. (a) MPEG-1 Audio Layer 3
   2. (b) Media Player 3
   3. (c) Music Protocol 3
   4. (d) Multi-Phase 3
3. In LibreOffice Calc, which symbol is used to start a formula?
   1. (a) #
   2. (b) @
   3. (c) =
   4. (d) +
4. Which type of chart is best for showing trends over time?
   1. (a) Pie Chart
   2. (b) Bar Chart
   3. (c) Line Chart
   4. (d) Area Chart
5. What is the function of the SUM formula in a spreadsheet?
   1. (a) Adds all numbers in a range
   2. (b) Subtracts numbers
   3. (c) Multiplies numbers
   4. (d) Finds the average

Section II: Short Answer Questions (5 — 2 = 10 marks)

Answer briefly:

1. Define multimedia. Name any three components of multimedia.

2. What is the difference between audio and video compression?
3. Write the syntax of the IF function in LibreOffice Calc with an example.
4. What is a database? Name any two database management systems.
5. What is the purpose of animation in multimedia presentations?

Section III: Long Answer Questions (1 — 5 = 5 marks)

Answer in detail:

1. Explain the steps to create a simple animation in Scratch. Describe the use of at least three different blocks in your animation.

Part B: Hands-on Practical Lab Task (5 marks)

Complete the following tasks on the computer:

1. Open LibreOffice Calc and create a marks sheet for 5 students with columns: Name, Maths, Science, English, Total, Average. Use the SUM and AVERAGE formulas. (2 marks)
2. Open Audacity, record a 10-second voice clip, and apply the "Noise Reduction" effect. Export the file as "MyVoice.mp3". (2 marks)
3. Save your spreadsheet file on the desktop with the filename "MarksSheet.ods". (1 mark)

Part C: Notebook Maintenance (5 marks)

Marks will be awarded for:

- Regularity in completing classwork (2 marks)
- Neatness of notes and diagrams (2 marks)
- Completion of all class activities (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters

Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 7 - Formative Assessment 3

Formative Assessment (FA-3)

- Class 7

Class: Class 7

Assessment Type: Formative Assessment (FA-3)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Programming - Scratch, Digital Citizenship

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 — 1 = 5 marks)

Choose the correct answer:

1. In Scratch, which block is used to create a copy of a sprite?
    1. (a) create clone of myself
    2. (b) delete this clone
    3. (c) when I start as a clone
    4. (d) switch costume to
2. What is a “digital footprint”?
    1. (a) A footprint made using digital tools
    2. (b) The trail of data left by online activities
    3. (c) A type of computer virus
    4. (d) A digital drawing tool
3. Which of the following is a cybercrime?
    1. (a) Sending a greeting card by email
    2. (b) Hacking into someone’s account
    3. (c) Searching information on Google
    4. (d) Using social media responsibly
4. In Scratch, what does the “forever” block do?
    1. (a) Runs the code only once
    2. (b) Runs the code repeatedly until stopped
    3. (c) Stops the program
    4. (d) Creates a new sprite
5. What is the primary purpose of a strong password?
    1. (a) To make the account look good
    2. (b) To protect the account from unauthorized access
    3. (c) To remember easily
    4. (d) To share with friends

Section II: Short Answer Questions (5 — 2 = 10 marks)

Answer briefly:

1. What are “clones” in Scratch? How are they created?

2. Name any two rules of cyber safety that every student should follow.
3. What is the difference between a “local variable” and a “global variable” in Scratch?
4. What is phishing? How can you identify a phishing email?
5. Why is it important to create a custom block in Scratch? Give one advantage.

Section III: Long Answer Questions (1 — 5 = 5 marks)

Answer in detail:

1. Describe the steps to create a simple “Catch the Falling Objects” game in Scratch. Explain the role of at least four different blocks used in the game.

Part B: Hands-on Practical Lab Task (5 marks)

Complete the following tasks on the computer:

1. Create a Scratch project where a cat sprite chases a mouse sprite using arrow keys. Use the “point towards” and “move” blocks. (2 marks)
2. Create a Scratch project that uses at least 3 clones to display multiple stars on the screen. Use the “create clone of” block. (2 marks)
3. Save your Scratch projects on the desktop with the filenames “ChaseGame.sb3” and “ClonesDemo.sb3”. (1 mark)

Part C: Notebook Maintenance (5 marks)

Marks will be awarded for:

- Regularity in completing classwork (2 marks)
- Neatness of notes and diagrams (2 marks)
- Completion of all class activities (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters

Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 7 - Formative Assessment 4

Formative Assessment (FA-4)

- Class 7

Class: Class 7

Assessment Type: Formative Assessment (FA-4)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Programming - Scratch, Digital Citizenship, Revision

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 — 1 = 5 marks)

Choose the correct answer:

1. Which Scratch block is used to play a sound when an event occurs?
   1. (a) play sound until done
   2. (b) stop all sounds
   3. (c) set volume to
   4. (d) change volume by
2. What does “CC” stand for when sharing images online?
   1. (a) Computer Code
   2. (b) Creative Commons
   3. (c) Cyber Crime
   4. (d) Computer Creative
3. In networking, what does the protocol HTTP stand for?
   1. (a) HyperText Transfer Protocol
   2. (b) High Text Transfer Protocol
   3. (c) HyperText Transmission Process
   4. (d) High Transfer Text Protocol
4. Which of the following is a correct way to protect your password?
   1. (a) Share it with close friends
   2. (b) Write it on your desk
   3. (c) Use a combination of letters, numbers, and symbols
   4. (d) Use the same password for all accounts
5. In GIMP, which file format supports layers?
   1. (a) JPEG
   2. (b) BMP
   3. (c) PNG
   4. (d) XCF

Section II: Short Answer Questions (5 — 2 = 10 marks)

Answer briefly:

1. What is the difference between “broadcast” and “broadcast and wait” blocks in Scratch?

2. Explain the concept of “digital footprint” and how it can affect your future.
3. What are the three types of network topologies? Name them.
4. Write any two uses of the AVERAGE function in LibreOffice Calc.
5. What is the difference between “Save” and “Save As” in any software?

Section III: Long Answer Questions (1 — 5 = 5 marks)

Answer in detail:

1. Explain the difference between LAN, MAN, and WAN with suitable diagrams and examples. Which type of network is used in a school?

Part B: Hands-on Practical Lab Task (5 marks)

Complete the following tasks on the computer:

1. Create a Scratch project that uses a custom block to draw a square shape using the pen extension. (2 marks)
2. Open a web browser, search for information on “cyber safety tips for students”, and write down five tips in a text file. (2 marks)
3. Save both files on the desktop with appropriate filenames. (1 mark)

Part C: Notebook Maintenance (5 marks)

Marks will be awarded for:

- Regularity in completing classwork (2 marks)
- Neatness of notes and diagrams (2 marks)
- Completion of all class activities (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters

Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 7 - Practical Lab Exam 1

Practical Lab Exam 1

- Class 7

Class: Class 7

Assessment Type: Practical Lab Exam 1

Total Marks: 20

Duration: 45 minutes

Topics Covered: Networking & Internet, Advanced Graphics, Multimedia, Data Handling

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Lab Tasks (15 marks)

Complete the following tasks on the computer:

Task 1: Network Identification (3 marks)

1. Open the Command Prompt on your computer.
2. Type `ipconfig` and press Enter.
3. Note down the IPv4 Address and Default Gateway.
4. Answer the following questions in a text file:
   - What is your IPv4 Address?
   - What is your Default Gateway?
   - What type of network are you connected to (LAN/WAN)?

Task 2: Image Editing in GIMP (4 marks)

1. Open GIMP and create a new image (600—400 pixels).
2. Create three layers: Background, Shape, and Text.
3. On the Background layer, fill with a gradient color.
4. On the Shape layer, draw a rectangle and a circle using the selection tools.
5. On the Text layer, type your name using the Text tool.
6. Save the file as “MyArt.xcf” on the desktop.

Task 3: Audio Recording in Audacity (3 marks)

1. Open Audacity.
2. Record a 15-second audio clip of your voice.
3. Apply the following effects:
   - Noise Reduction
   - Amplify
4. Export the audio as “MyVoice.mp3” on the desktop.

Task 4: Spreadsheet in LibreOffice Calc (5 marks)

1. Open LibreOffice Calc and create a table with the following data:

   | Student Name | Maths | Science | English | Total | Average | |————-|——-|———|———|——-|———| | Rahul | 85 | 78 | 90 | | | | Priya | 92 | 88 | 85 | | | | Arjun | 78 | 92 | 88 | | |
2. Use the SUM formula to calculate Total for each student.
3. Use the AVERAGE formula to calculate Average for each student.

4. Create a bar chart comparing the three students' marks.
5. Save the file as "MarksAnalysis.ods" on the desktop.

Viva Voce (5 marks)

Answer the following questions verbally:

1. What is the difference between LAN and WAN? (1 mark)
2. Why are layers important in image editing? (1 mark)
3. What is the use of the SUM function in a spreadsheet? (1 mark)
4. Name any two audio formats. (1 mark)
5. What is the purpose of a firewall in a network? (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters
Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)

- Real-life connections: “Where I see this outside school” (1 mark)

ICT Class 7 - Practical Lab Exam 2

Practical Lab Exam 2

- Class 7

Class: Class 7

Assessment Type: Practical Lab Exam 2

Total Marks: 20

Duration: 45 minutes

Topics Covered: Complete Syllabus - All Practical Activities

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Lab Tasks (15 marks)

Complete the following tasks on the computer:

Task 1: Scratch Game Development (5 marks)

Create a "Catch the Falling Stars" game in Scratch with the following requirements:

1. Create a cat sprite that moves left and right using the left and right arrow keys. (1 mark)
2. Create a star sprite that falls from the top of the screen randomly. (1 mark)
3. When the cat catches the star, the score should increase by 1. (1 mark)
4. When the star touches the bottom edge, the game should display "Game Over". (1 mark)
5. Save the project as "CatchStars.sb3" on the desktop. (1 mark)

Task 2: Advanced GIMP Editing (4 marks)

1. Open any image from the desktop or Internet in GIMP.
2. Perform the following operations:
   - Duplicate the image and work on the copy.
   - Apply the Gaussian Blur filter to the background.
   - Use the Fuzzy Select tool to select and remove the background of an object.
   - Add a text overlay with your name in a different color.
3. Save the edited image as "EditedPhoto.xcf" on the desktop.

Task 3: Data Analysis in LibreOffice Calc (4 marks)

1. Open LibreOffice Calc and create a student database with the following columns:
   - Roll No, Name, Class, Marks Obtained, Grade, Status
2. Enter data for at least 5 students.
3. Use the following functions:
   - IF function to assign Grade (A for 90+, B for 75-89, C for 60-74, D for below 60).
   - IF function to assign Status (Pass if marks >= 35, Fail if below 35).
4. Create a pie chart showing the grade distribution.
5. Save the file as "StudentDatabase.ods" on the desktop.

Task 4: Internet Research and Documentation (2 marks)

1. Open a web browser and research "Online Safety Tips for Students".
2. Create a text file with the following:
   - Title: "Online Safety Guide"

  - At least 5 safety tips with brief explanations.

3. Save the file as “SafetyGuide.txt” on the desktop.

Viva Voce (5 marks)

Answer the following questions verbally:

1. What are clones in Scratch? How are they created? (1 mark)
2. Explain the difference between “Save” and “Save As” in any software. (1 mark)
3. What is the IF function? Give a practical example of its use. (1 mark)
4. What are the ethical considerations when using images from the Internet? (1 mark)
5. Explain the concept of “digital footprint” with a real-life example. (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters
Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others’ solutions

Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: “What I learned” and “What was hard” (2 marks)

- Real-life connections: “Where I see this outside school” (1 mark)

ICT Class 7 - Summative Assessment 1

Summative Assessment (SA-1)

- Class 7

Class: Class 7

Assessment Type: Summative Assessment (SA-1)

Total Marks: 40

Duration: 2 hours

Topics Covered: Ch 1: Networking & Internet, Ch 2: Advanced Graphics, Ch 3: Multimedia, Ch 4: Data Handling

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Multiple Choice Questions (10 — 1 = 10 marks)

Choose the correct answer:

1. A network that spans across a city is called:
    1. (a) LAN
    2. (b) WAN
    3. (c) MAN
    4. (d) PAN
2. Which protocol is used for secure web browsing?
    1. (a) HTTP
    2. (b) HTTPS
    3. (c) FTP
    4. (d) SMTP
3. The full form of URL is:
    1. (a) Uniform Resource Locator
    2. (b) Universal Resource Locator
    3. (c) Uniform Resource Link
    4. (d) Universal Resource Link
4. In GIMP, which tool is used to select a rectangular area?
    1. (a) Ellipse Select Tool
    2. (b) Rectangle Select Tool
    3. (c) Free Select Tool
    4. (d) Scissors Select Tool
5. Which audio format is commonly used for music files?
    1. (a) .txt
    2. (b) .doc
    3. (c) .mp3
    4. (d) .xlsx
6. In LibreOffice Calc, the formula =SUM(A1:A5) will:
    1. (a) Add values from A1 to A5
    2. (b) Subtract values from A1 to A5
    3. (c) Find the average of A1 to A5
    4. (d) Count cells from A1 to A5

7. Which of the following is a vector graphics software?
    1. (a) GIMP
    2. (b) Inkscape
    3. (c) Audacity
    4. (d) VLC
8. What is the purpose of a firewall?
    1. (a) To increase Internet speed
    2. (b) To protect a network from unauthorized access
    3. (c) To edit images
    4. (d) To play videos
9. In a spreadsheet, what does the “AutoSum” feature do?
    1. (a) Automatically deletes data
    2. (b) Automatically calculates the sum of selected cells
    3. (c) Automatically formats the cells
    4. (d) Automatically prints the data
10. Which multimedia component deals with moving pictures?
    1. (a) Text
    2. (b) Audio
    3. (c) Video
    4. (d) Graphics

Part B: Short Answer Questions (8 — 2 = 16 marks)

Answer briefly:

1. Differentiate between LAN and WAN with one example each.
2. What is a search engine? Name any two search engines.
3. Explain the use of layers in GIMP. Why are they important for image editing?
4. What is the difference between raster and vector graphics? Give one example of each.
5. Define multimedia. List its five main components.
6. Write the syntax of the IF function in LibreOffice Calc with a practical example.
7. What is the difference between a formula and a function in a spreadsheet?
8. What are the advantages of using charts in data analysis?

Part C: Long Answer Questions (4 — 3 = 12 marks)

Answer in detail:

1. Explain the types of computer networks (LAN, MAN, WAN) with a neat diagram. Compare their features, advantages, and disadvantages.
2. Describe the steps to edit an image in GIMP using layers. Explain how to add a new layer, draw on it, and merge layers.

3. What is multimedia? Explain the role of audio and video in multimedia with examples. How is compression used in multimedia?
4. Explain the components of a spreadsheet with a suitable example. Describe how to use SUM, AVERAGE, MAX, and MIN functions.

Part D: Application Based Questions (2 — 1 = 2 marks)

1. Raj wants to create a photo album with multiple images. Which image editing software would you recommend and why? Mention two features that would help him.
2. A school wants to connect computers in 5 different buildings across a campus. Which type of network would be most suitable? Justify your answer.

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters
Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)

- Real-life connections: “Where I see this outside school” (1 mark)

ICT Class 7 - Summative Assessment 2

Summative Assessment (SA-2)

- Class 7

Class: Class 7

Assessment Type: Summative Assessment (SA-2)

Total Marks: 40

Duration: 2 hours

Topics Covered: Ch 1-6: Complete Syllabus

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Multiple Choice Questions (10 — 1 = 10 marks)

Choose the correct answer:

1. Which Scratch block category contains the “play sound” block?
    1. (a) Motion
    2. (b) Looks
    3. (c) Sound
    4. (d) Events
2. What is the full form of “CC” in licensing?
    1. (a) Computer Code
    2. (b) Creative Commons
    3. (c) Cyber Crime
    4. (d) Copyright Content
3. Which protocol is used to receive emails?
    1. (a) SMTP
    2. (b) POP3
    3. (c) HTTP
    4. (d) FTP
4. In GIMP, which filter is used to blur an image?
    1. (a) Sharpen
    2. (b) Gaussian Blur
    3. (c) Brightness-Contrast
    4. (d) Color Balance
5. The SUM function in LibreOffice Calc is used to:
    1. (a) Find the total of a range of cells
    2. (b) Count the number of cells
    3. (c) Find the largest value
    4. (d) Sort data
6. What is a “digital footprint”?
    1. (a) A footprint made on a digital screen
    2. (b) The trail of data left by a user’s online activities
    3. (c) A type of malware
    4. (d) A digital drawing tool

7. Which block is used to create a clone of a sprite in Scratch?
    1. (a) create clone of myself
    2. (b) delete this clone
    3. (c) when I start as a clone
    4. (d) forever
8. Which of the following is NOT a type of network topology?
    1. (a) Star
    2. (b) Bus
    3. (c) Ring
    4. (d) Tree Topology
9. What is the purpose of the “stop all” block in Scratch?
    1. (a) Stops only the current script
    2. (b) Stops all scripts in the project
    3. (c) Pauses the program
    4. (d) Deletes the sprite
10. Which of the following is a safe online practice?
    1. (a) Sharing your password with friends
    2. (b) Clicking on unknown links in emails
    3. (c) Keeping software and antivirus updated
    4. (d) Downloading files from untrusted websites

Part B: Short Answer Questions (8 — 2 = 16 marks)

Answer briefly:

1. What is the difference between a client and a server in a network?
2. Explain any two Internet services with their uses.
3. What are layers in GIMP? How do they help in editing images?
4. What is the difference between JPEG and PNG image formats?
5. Explain the concept of animation. What are the two main types of animation?
6. What is a formula in a spreadsheet? Write the syntax of the AVERAGE function.
7. What are “clones” in Scratch? How are they useful in game development?
8. What is plagiarism? Why should students avoid it?

Part C: Long Answer Questions (4 — 3 = 12 marks)

Answer in detail:

1. Explain the complete anatomy of a computer network with a diagram. Include the role of routers, switches, and modems in a network.
2. Describe the steps to create an animated story in Scratch. Include the use of at least four different block categories.

3. What is data handling? Explain the steps to create a dashboard in LibreOffice Calc using charts and formulas. Give a practical example.
4. Explain the importance of digital citizenship. Describe any five rules of responsible technology use with examples.

Part D: Application Based Questions (2 — 1 = 2 marks)

1. Meera wants to create a presentation with background music and animated images. Which multimedia tools would you recommend? Explain how she can use them.
2. A student receives an email asking for their bank details with a link to “update account information”. What should the student do and why?

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters
Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others’ solutions

Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: “What I learned” and “What was hard” (2 marks)

- Real-life connections: “Where I see this outside school” (1 mark)

ICT Class 7 - Unit Test 1

Unit Test 1

- Class 7

Class: Class 7

Assessment Type: Unit Test 1

Total Marks: 10

Duration: 30 minutes

Topics Covered: Networking & Internet

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Section I: Multiple Choice Questions (5 — 1 = 5 marks)

Choose the correct answer:

1. A network that covers a small geographical area like a single building is called:
    1. (a) WAN
    2. (b) LAN
    3. (c) MAN
    4. (d) SAN
2. Which of the following is NOT an Internet service?
    1. (a) Email
    2. (b) FTP
    3. (c) MS Word
    4. (d) WWW
3. The full form of ISP is:
    1. (a) Internet Service Provider
    2. (b) Internal Service Protocol
    3. (c) Information System Provider
    4. (d) Internet System Protocol
4. Which protocol is used to transfer files between computers over a network?
    1. (a) HTTP
    2. (b) SMTP
    3. (c) FTP
    4. (d) TCP
5. A MAN stands for:
    1. (a) Metropolitan Area Network
    2. (b) Metropolitan Access Network
    3. (c) Modified Area Network
    4. (d) Multi Access Network

Section II: Short Answer Questions (5 — 1 = 5 marks)

Answer briefly:

1. What is a computer network? State its two advantages.

2. Differentiate between client and server in a network.
3. What is the role of a modem in a computer network?
4. Name any two web browsers and one search engine.
5. What is an IP address? Why is it important?

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters
Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: ***Date:***
Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: “What I learned” and “What was hard” (2 marks)
- Real-life connections: “Where I see this outside school” (1 mark)

ICT Class 7 - Unit Test 2

Unit Test 2

- Class 7

Class: Class 7

Assessment Type: Unit Test 2

Total Marks: 10

Duration: 30 minutes

Topics Covered: Multimedia

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Section I: Multiple Choice Questions (5 — 1 = 5 marks)

Choose the correct answer:

1. Which of the following is an audio editing software?
   1. (a) GIMP
   2. (b) Audacity
   3. (c) Scratch
   4. (d) LibreOffice Calc
2. What is the full form of MIDI?
   1. (a) Musical Instrument Digital Interface
   2. (b) Media Instrument Digital Interface
   3. (c) Musical Input Device Interface
   4. (d) Media Input Device Interface
3. The file extension for an MP4 video file is:
   1. (a) .mp4
   2. (b) .mp3
   3. (c) .wav
   4. (d) .avi
4. Which multimedia element involves moving images with sound?
   1. (a) Text
   2. (b) Audio
   3. (c) Video
   4. (d) Graphics
5. Compression of multimedia files is done to:
   1. (a) Increase file size
   2. (b) Reduce file size
   3. (c) Delete the file
   4. (d) Change file format

Section II: Short Answer Questions (5 — 1 = 5 marks)

Answer briefly:

1. What is multimedia? List its five components.

2. What is the difference between WAV and MP3 audio formats?
3. Define animation. Give one real-life example of animation.
4. What is the purpose of a media player?
5. What are the uses of multimedia in education?

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters
Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 7 - Unit Test 3

Unit Test 3

- Class 7

Class: Class 7

Assessment Type: Unit Test 3

Total Marks: 10

Duration: 30 minutes

Topics Covered: Programming - Scratch

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Section I: Multiple Choice Questions (5 — 1 = 5 marks)

Choose the correct answer:

1. In Scratch, which block category contains the "say" and "think" blocks?
   1. (a) Motion
   2. (b) Looks
   3. (c) Sound
   4. (d) Events
2. What is the default sprite in a new Scratch project?
   1. (a) Dog
   2. (b) Cat
   3. (c) Bird
   4. (d) Fish
3. Which block is used to repeat a set of instructions a specific number of times?
   1. (a) forever
   2. (b) repeat 10
   3. (c) wait 1 seconds
   4. (d) stop all
4. To create a clone of a sprite, which block is used?
   1. (a) create clone of [myself]
   2. (b) delete this clone
   3. (c) when I start as a clone
   4. (d) switch costume to
5. Which event block is used to start a Scratch program when the green flag is clicked?
   1. (a) when [space] key pressed
   2. (b) when green flag clicked
   3. (c) when this sprite clicked
   4. (d) when I receive [message]

Section II: Short Answer Questions (5 — 1 = 5 marks)

Answer briefly:

1. What is Scratch? Who developed it?

2. Explain the difference between “Motion” and “Looks” blocks with one example each.
3. What is a “loop” in programming? Name two types of loops available in Scratch.
4. How do you create a custom block in Scratch? Write the steps.
5. What is the purpose of the “pen” extension in Scratch?

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters
Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others’ solutions

Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: “What I learned” and “What was hard” (2 marks)
- Real-life connections: “Where I see this outside school” (1 mark)

ICT Class 7 - Unit Test 4

Unit Test 4

- Class 7

Class: Class 7

Assessment Type: Unit Test 4

Total Marks: 10

Duration: 30 minutes

Topics Covered: Digital Citizenship

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Section I: Multiple Choice Questions (5 — 1 = 5 marks)

Choose the correct answer:

1. Which of the following is an example of cyberbullying?
    1. (a) Sending a friendly email
    2. (b) Posting hurtful comments about someone online
    3. (c) Sharing a funny video with friends
    4. (d) Using the Internet for research
2. A strong password should contain:
    1. (a) Only lowercase letters
    2. (b) Only numbers
    3. (c) A mix of uppercase, lowercase, numbers, and symbols
    4. (d) Your date of birth
3. What is “phishing”?
    1. (a) A method of catching fish
    2. (b) A fraudulent attempt to obtain sensitive information
    3. (c) A type of computer virus
    4. (d) A social media platform
4. Which of the following protects your computer from viruses?
    1. (a) Firewall
    2. (b) Antivirus software
    3. (c) Both (a) and (b)
    4. (d) None of these
5. Copyright means:
    1. (a) Anyone can copy any work
    2. (b) The creator has exclusive rights over their original work
    3. (c) Works cannot be shared at all
    4. (d) Only government can use creative works

Section II: Short Answer Questions (5 — 1 = 5 marks)

Answer briefly:

1. What is cyber safety? Why is it important for students?

2. Explain the term “digital footprint” with an example.
3. What are the rules of netiquette? State any three.
4. What is plagiarism? How can it be avoided?
5. Name any two ways to protect personal information online.

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters
Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others’ solutions

Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: “What I learned” and “What was hard” (2 marks)
- Real-life connections: “Where I see this outside school” (1 mark)